3.3 An algorithm is a plan

In this lesson

You will learn:

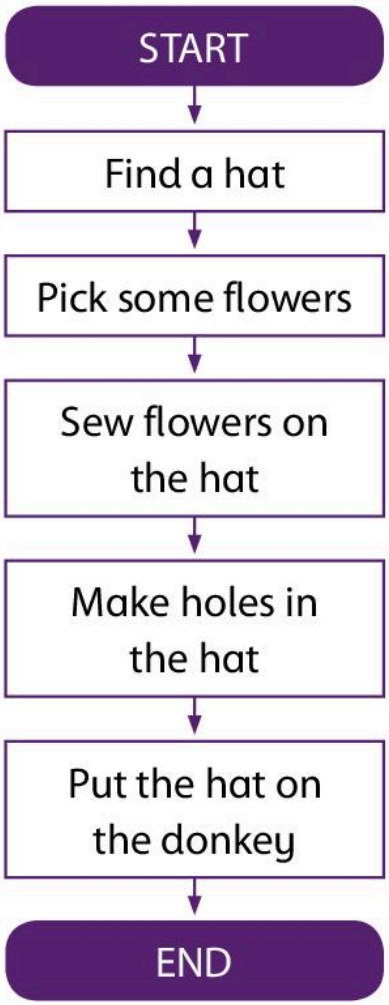
➔ what an algorithm is.

A plan that tells you how to do a task is called an **algorithm**.

The algorithm must include all the actions. The actions must be in the right order.

You can draw an algorithm. Put the actions in boxes. Join the boxes with arrows to show the right order.

Here is an algorithm for making a donkey's hat.



Activity



A big brother made a toy horse for his little sister. Here are the actions.

Find some wood



Carve the wood into a horse shape



Let the paint dry

Paint the horse



Buy some paint

Write the actions in the right order.

Extra challenge

Make an algorithm for making a toy horse. Write the actions in boxes joined with arrows. On a big piece of paper make a colourful picture.

Think again

What does 'algorithm' mean? Write or say the answer in your own words.

3.4 Algorithms and programs

In this lesson

You will learn:

- how making an algorithm helps you to plan a program.



Program

A **program** is a series of commands. The commands tell the computer what to do. When you **run a program** the computer carries out all the commands in the program.

A **programmer** is the person who makes a computer program.

A programmer sometimes makes an algorithm before they make the program. The algorithm is the plan for the program.

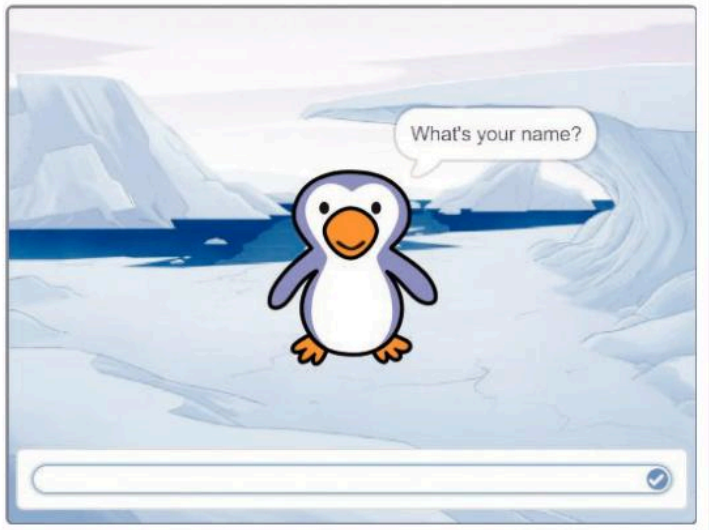
Sprite

You can make a program. You use a programming language to make programs. Scratch is a programming language.

- A Scratch program controls a **sprite**. A sprite is an object or creature on the screen. The program makes it do different things.
- A Scratch program is also called a **script**. A script is a short program.

Activity

A programmer wanted to make a program. The sprite is a penguin. Here is a list of things that the penguin will do.



Say goodbye

Go off the screen

Say hello and your name

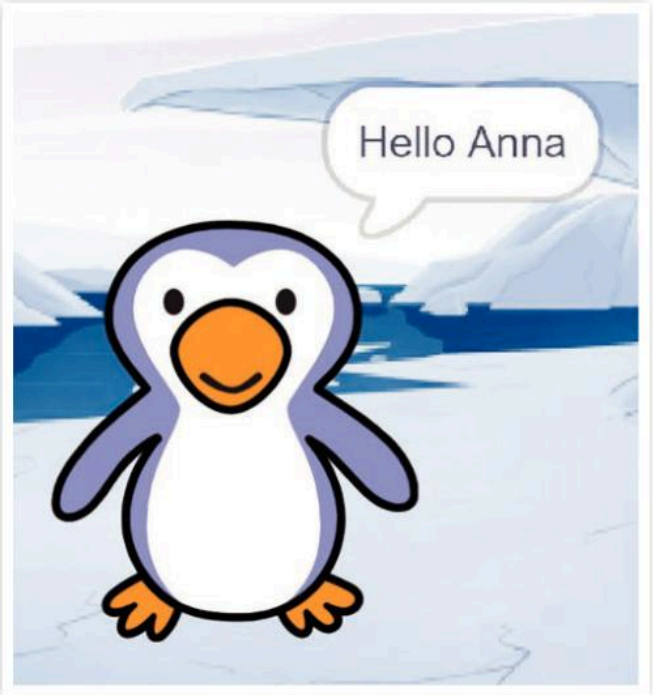
Ask your name

Come onto the screen

Think again

Write an algorithm showing the actions of the program in boxes joined by arrows.

Put these actions in the right order.



Extra challenge

Think of one extra action the penguin could do. Make a plan for the program that includes the extra action.

Be creative

Plan a new computer program:

- Draw the sprite.
- Say what actions the sprite will do.